

DBMS – ORDER AND ORDER ITEMS MANAGEMENT

Project Title: Big Basket E-commerce Database Management System

1. Database Design

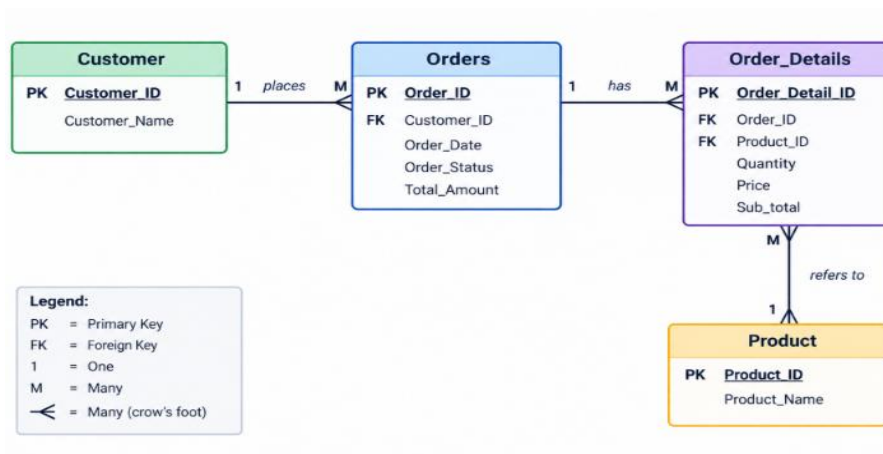
1.1 Order Table

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_date	-
Order_status	Check
Total_amount	-

1.2 Order_items Table

Attribute	Key Type
Order_item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Price	-

2. Primary and Foreign Key Relationship



3. Table Creation

3.1 Create Orders Table

```
CREATE TABLE Orders (  
    Order_ID NUMBER PRIMARY KEY,  
    Customer_ID NUMBER NOT NULL,  
    Order_Date DATE NOT NULL,  
    Order_Status VARCHAR2(20),  
    Total_Amount NUMBER(10,2),  
    CONSTRAINT fk_order_customer  
        FOREIGN KEY (Customer_ID)  
        REFERENCES Customer(Customer_ID),  
    CONSTRAINT chk_order_status  
        CHECK (Order_Status IN  
            ('Pending', 'Confirmed', 'Shipped', 'Delivered', 'Cancelled'))  
);
```

3.2 Create Order_Details Table

```
CREATE TABLE Order_Details (  
    Order_Detail_ID NUMBER PRIMARY KEY,  
    Order_ID NUMBER NOT NULL,  
    Product_ID NUMBER NOT NULL,  
    Quantity NUMBER NOT NULL,  
    Price NUMBER(10,2) NOT NULL,  
    Sub_total NUMBER(10,2),  
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),  
    FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
);
```

4. Data Insertion

4.1 Orders Records

```
INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1001, 101, TO_DATE('21-09-2026', 'DD-MM-YYYY'), 'Pending', 900);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1002, 102, TO_DATE('20-09-2026', 'DD-MM-YYYY'), 'Confirmed', 1250);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1003, 103, TO_DATE('19-09-2026', 'DD-MM-YYYY'), 'Shipped', 1800);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1004, 101, TO_DATE('18-09-2026', 'DD-MM-YYYY'), 'Delivered', 750);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1005, 104, TO_DATE('17-09-2026', 'DD-MM-YYYY'), 'Cancelled', 500);
1 row created.
```

```
INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1006, 105, TO_DATE('16-09-2026', 'DD-MM-YYYY'), 'Pending', 850);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1007, 106, TO_DATE('15-09-2026', 'DD-MM-YYYY'), 'Cancelled', 1200);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1008, 101, TO_DATE('04-09-2026', 'DD-MM-YYYY'), 'Delivered', 1500);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1009, 107, TO_DATE('09-09-2026', 'DD-MM-YYYY'), 'Shipped', 2500);
1 row created.

INSERT INTO Orders
(Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(1010, 109, TO_DATE('20-09-2026', 'DD-MM-YYYY'), 'Confirmed', 28000);
1 row created.
```

4.2 Order_Details Records

```
INSERT INTO Order_Details
VALUES (1, 1001, 101, 1, 70000, 70000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (2, 1002, 104, 1, 5000, 5000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (3, 1003, 102, 1, 45000, 45000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (4, 1004, 103, 2, 9000, 18000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (5, 1005, 106, 1, 12500, 12500);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (6, 1006, 107, 2, 16000, 32000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (7, 1007, 108, 1, 8500, 8500);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (8, 1008, 105, 2, 11000, 22000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (9, 1009, 109, 1, 15000, 15000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (10, 1010, 110, 2, 14000, 28000);
1 row created.
```

```
INSERT INTO Order_Details
VALUES (11, 1001, 104, 2, 5250, 10500);
1 row created.
```

5. Data Retrieval

5.1 Display Orders

```
Select * From Orders;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	ORDER_STATUS	TOTAL_AMOUNT
1001	101	21-SEP-26	Pending	900
1002	102	20-SEP-26	Confirmed	1250
1003	103	19-SEP-26	Shipped	1800
1004	101	18-SEP-26	Delivered	750
1005	104	17-SEP-26	Cancelled	500
1006	105	16-SEP-26	Pending	850
1007	106	15-SEP-26	Cancelled	1200
1008	101	04-SEP-26	Delivered	1500
1009	107	09-SEP-26	Shipped	2500
1010	109	20-SEP-26	Confirmed	28000

```
10 rows selected.
```

5.2 Display Order_Details

```
Select * from Order_Details;
```

ORDER_DETAIL_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE	SUB_TOTAL
1	1001	101	1	70000	70000
2	1002	104	1	5000	5000
3	1003	102	1	45000	45000
4	1004	103	2	9000	18000
5	1005	106	1	12500	12500
6	1006	107	2	16000	32000
7	1007	108	1	8500	8500
8	1008	105	2	11000	22000
9	1009	109	1	15000	15000
10	1010	110	2	14000	28000
11	1001	104	2	5250	10500

```
11 rows selected.
```

6. Order and Order_items Modification

6.1 Update Order Status

```
UPDATE Orders
SET Order_Status = 'Confirmed'
WHERE Order_ID = 1001;
1 row updated.
```

```
UPDATE Orders
SET Order_Status = 'Shipped'
WHERE Order_ID = 1002;
1 row updated
```

```
Select *
from Orders
Where Order_ID in(1001,1002);
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	ORDER_STATUS	TOTAL_AMOUNT
1001	101	21-SEP-26	Confirmed	900
1002	102	20-SEP-26	Confirmed	1250

6.2 Update Order Quantity

```
UPDATE Order_Details
SET Quantity = 2
WHERE Order_Detail_ID = 1;
1 row updated.
```

```
Update Order_Details
SET price=price+100
Where Order_Detail_ID=3;
1 row updated.
```

```
Update Order_Details
SET Quantity=2
Where Order_Detail_ID=10;
1 row updated.
```

```
Select *
from Order_Details
Where Order_Detail_ID in(1,3,10);
```

ORDER_DETAIL_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE	SUB_TOTAL
1	1001	101	2	70000	70000
3	1003	102	1	45100	45000
10	1010	110	2	14000	28000

```
commit;
```

7. Customer-Order History Report:

The output displays each customer's order history, including order ID, order date, product ID, quantity, price, and subtotal. It helps track the products purchased by each customer and the details of their orders.

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    o.Order_ID,
    o.Order_Date,
    od.Product_ID,
    od.Quantity,
    od.Price,
    od.Sub_total
FROM Customer c
JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
JOIN Order_Details od
    ON o.Order_ID = od.Order_ID
ORDER BY c.Customer_ID, o.Order_Date;
```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
101	Arun Kumar	1008
04-SEP-26	105	2
		11000
		22000
101	Arun Kumar	1004
18-SEP-26	103	2
		9000
		18000
101	Arun Kumar	1001
21-SEP-26	101	2
		70000
		70000

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
101	Arun Kumar	1001
21-SEP-26	104	2
		5250
		10500
102	Priya Sharma	1002
20-SEP-26	104	1
		5000
		5000
103	Rahul Singh	1003
19-SEP-26	102	1
		45100
		45000

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
104	Sneha Reddy	1005
17-SEP-26	106	1
	12500	12500
105	Vikram Patel	1006
16-SEP-26	107	2
	16000	32000
106	Anjali Das	1007
15-SEP-26	108	1
	8500	8500

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
107	Karan Mehta	1009
09-SEP-26	109	1
	15000	15000
109	Rohan Gupta	1010
20-SEP-26	110	2
	14000	28000

9 rows selected.

8. Customer -wise Order Summary:

The output shows each customer along with the **number of orders placed and the total amount spent**. It summarises each customer's ordering activity and total purchase value.

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    COUNT(o.Order_ID) AS Total_Orders,
    SUM(o.Total_Amount) AS Total_Amount
FROM Customer c
JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
GROUP BY
    c.Customer_ID,
    c.Customer_Name
ORDER BY
    c.Customer_ID;
```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS
-------------	---------------	--------------

TOTAL_AMOUNT

101	Arun Kumar	3
3150		
102	Priya Sharma	1
1250		
103	Rahul Singh	1
1800		

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS
-------------	---------------	--------------

TOTAL_AMOUNT

104	Sneha Reddy	1
500		
105	Vikram Patel	1
850		
106	Anjali Das	1
1200		

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS
-------------	---------------	--------------

TOTAL_AMOUNT

107	Karan Mehta	1
2500		
109	Rohan Gupta	1
28000		

8 rows selected.

commit;

Commit Completed.